A-A-624A
December 7, 1987
SUPERSEDING
A-A-624
November 23, 1979

COMMERCIAL ITEM DESCRIPTION

TAP, THREAD CUTTING
(REGULAR, MACHINE SCREW SIZES)

The General Services Administration has authorized the use of this commercial item description in preference to Type I of Federal Specification GGG-T-70.

This commercial item description covers general purpose, ground thread, high-speed steel taps with straight flutes for use in hand and machine tapping operations.

Salient characteristics:

Design. Taps shall be of the regular hand type with straight flutes, and be in accordance with the latest revision of American National Standard ANSI B94.9.

Performance. Taps shall be capable of cutting smooth, well-defined UN Form threads.

Material. Material shall be high-speed steel of the same grade used for taps supplied to the commercial market for general purpose applications.

Hardness. Taps shall be hardened and tempered so as to indicate a Rockwell C62-66 hardness in at least eighty percent of the threaded section from the threaded end. The shank and square drive end may have other suitable heat treatment characteristics to prevent twisting and breaking of the tap.

Finish. Threads shall have a ground finish. Other areas shall be finished in accordance with commercial practice. Coatings or surface treatments, if provided, shall be suitable for general purpose tapping applications.

Thread size, type of chamfer, pitch diameter, and number of flutes. Shall be as shown in the following table for the National Stock Number (NSN) specified. Except as indicated, threads shall be right hand direction.

NATIONAL STOCK NUMBER	SIZE, SERIES, DIRECTION	CHAMFER TYPE	P.D. LIM. SYM.	PITCH MIN. (INCH)	DIAMETER MAX. (INCH)	NUMBER OF FLUTES
5136-00-189-7812	NO. 1-64UNC	BOTTOMING	H1	0.0629	0.0634	2
5136-00-189-7814	NO. 1-64UNC	PLUG	H1	0.0629	0.0634	2
5136-00-189-7816	NO. 1-64UNC	TAPER	H1	0.0629	0.0634	2
5136-00-189-7818	NO. 0-80UNF	BOTTOMING	H1	0.0519	0.0524	2
5136-00-189-7819	NO. 1-72UNF	BOTTOMING	H1	0.0640	0.0645	2
5136-00-189-7824	NO. 0-80UNF	TAPER	H1	0.0519	0.0524	2
5136-00-189-7825	NO. 1-72UNF	TAPER	H1	0.0640	0.0645	2
5136-00-189-7826	NO. 2-64UNF	TAPER	H1*	0.0759	0.0764	3
5136-00-203-6212	NO. 10-32UNF	BOTTOMING	H1	0.1697	0.1702	4
5136-00-203-6217	NO. 10-32UNFLH	PLUG	H1	0.1697	0.1702	4
5136-00-203-6500	NO. 8-32UNC	BOTTOMING	H1	0.1437	0.1442	4
5136-00-203-6501	NO. 10-24UNC	BOTTOMING	H1	0.1629	0.1634	4
5136-00-203-6502	NO. 12-24UNC	BOTTOMING	H1*	0.1889	0.1894	4
5136-00-203-6506	NO. 8-32UNC	PLUG	H1	0.1437	0.1442	4
5136-00-203-6508	NO. 10-24UNC	PLUG	H1	0.1629	0.1634	4
5136-00-203-7074	NO. 8-32UNC	TAPER	H1	0.1437	0.1442	4
5136-00-204-1644	NO. 6-32UNC	TAPER	H1	0.1177	0.1182	3
5136-00-221-7874	NO. 12-24UNC	PLUG	H2*	0.1894	0.1899	4
5136-00-227-9455	NO. 6-32UNC	BOTTOMING	H1	0.1177	0.1182	3
5136-00-228-0989	NO. 5-44UNF	TAPER	H2	0.1107	0.1112	3
5136-00-228-1004	NO. 4-48UNF	PLUG	H2	0.0990	0.0995	3
5136-00-228-1005	NO. 5-44UNF	PLUG	H2	0.1107	0.1112	3

FSC 5136

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:09Z
Check the source to verify that this is the current version before use.

NATIONAL STOCK NUMBER	SIZE, SERIES, DIRECTION	CHAMFER TYPE	P.D. LIM. SYM.	PITCH MIN. (INCH)	DIAMETER MAX. (INCH)	NUMBER OF FLUTES
5136-00-228-1006	NO. 6-40UNF	PLUG	H2	0.1223	0.1228	3
5136-00-228-1007	NO. 8-36UNF	PLUG	H2	0.1465	0.1470	4
5136-00-228-1008	NO. 10-32UNF	PLUG	H2	0.1702	0.1707	4
5136-00-228-1009	NO. 12-28UNF	PLUG	H2*	0.1933	0.1938	4
5136-00-228-1039	NO. 3-48UNC	PLUG	H2	0.0860	0.0865	3
5136-00-228-1040	NO. 4-40UNC	PLUG	H2	0.0963	0.0968	3
5136-00-232-1974	NO. 8-32UNC	BOTTOMING	H2	0.1442	0.1447	4
5136-00-262-1484	NO. 2-64UNF	BOTTOMING	H2	0.0764	0.0769	3
5136-00-262-3945	NO. 0-80UNF	PLUG	H2	0.0524	0.0529	2
5136-00-272-9760	NO. 0-80UNF	PLUG	H1	0.0519	0.0524	2
5136-00-272-9761	NO. 1-72UNF	PLUG	H1	0.0640	0.0645	2
5136-00-272-9762	NO. 3-56UNF	PLUG	H1*	0.0874	0.0879	3
5136-00-516-2948	NO. 6-40UNF	PLUG	H1	0.1218	0.1223	3
5136-00-516-2950	NO. 8-36UNF	PLUG	H1*	0.1460	0.1465	4
5136-00-516-3370	NO. 10-32UNF	TAPER	H1	0.1697	0.1702	4
5136-00-516-3373	NO. 6-32UNC	PLUG	H1	0.1177	0.1182	3
5136-00-555-8906	NO. 10-32UNF	BOTTOMING	H2	0.1702	0.1707	4
5136-00-580-7152	NO. 3-48UNC	BOTTOMING	H2	0.0860	0.0865	3
5136-00-580-7154	NO. 5-44UNF	BOTTOMING	H2	0.1107	0.1112	3
5136-00-580-7155	NO. 12-28UNF	BOTTOMING	H2*	0.1933	0.1938	4
5136-00-580-7316	NO. 6-32UNC	BOTTOMING	H2	0.1182	0.1187	3
5136-00-580-7323	NO. 3-56UNF	PLUG	H2	0.0879	0.0884	3
5136-00-580-7327	NO. 10-24UNC	BOTTOMING	H2	0.1634	0.1639	4
5136-00-580-7330	NO. 4-48UNF	BOTTOMING	H2	0.0990	0.0995	3
5136-00-580-7331	NO. 6-40UNF	BOTTOMING	H2	0.1223	0.1228	3
5136-00-580-7332	NO. 8-36UNF	BOTTOMING	H2	0.1465	0.1470	4
5136-00-580-7351	NO. 5-40UNC	BOTTOMING	H2	0.1093	0.1098	3
5136-00-580-7357	NO. 12-24UNC	BOTTOMING	H2*	0.1894	0.1899	4
5136-00-580-7361	NO. 5-40UNC	PLUG	H2	0.1093	0.1098	3
5136-00-585-6760	NO. 10-24UNC	PLUG	H2	0.1634	0.1639	4
5136-00-729-5675	NO. 6-40UNF	TAPER	H2	0.1223	0.1228	3
5136-00-729-5677	NO. 3-56UNF	TAPER	H2	0.0879	0.0884	3
5136-00-729-5678	NO. 4-48UNF	TAPER	H2	0.0990	0.0995	3
5136-00-729-5684	NO. 10-24UNC	TAPER	H2	0.1634	0.1639	4
5136-00-729-5685	NO. 5-40UNC	TAPER	H2	0.1093	0.1098	3
5136-00-729-5686	NO. 4-40UNC	TAPER	H2	0.0963	0.0968	3
5136-00-729-5687	NO. 3-48UNC	TAPER	H2	0.0860	0.0865	3
5136-00-729-5689	NO. 4-40UNC	BOTTOMING	H2	0.0963	0.0968	3
5136-00-729-5694	NO. 8-32UNC	PLUG	H2	0.1442	0.1447	4
5136-00-729-5695	NO. 6-32UNC	PLUG	H2	0.1182	0.1187	3
5136-00-729-5698	NO. 8-36UNF	TAPER	H2	0.1465	0.1470	4
5136-00-729-5700	NO. 10-32UNF	PLUG	H1	0.1697	0.1702	4
5136-00-729-5711	NO. 3-56UNF	BOTTOMING	H2	0.0879	0.0884	3
5136-00-820-8167	NO. 2-56UNC	TAPER	H2	0.0749	0.0754	3
5136-00-821-1026	NO. 4-36UNS	TAPER	H2	0.0945	0.0950	3
5136-00-821-1027	NO. 6-32UNC	TAPER	H2	0.1182	0.1187	3
5136-00-821-1029	NO. 10-32UNF	TAPER	H2	0.1702	0.1707	4
5136-00-821-1030	NO. 12-24UNC	TAPER	H2*	0.1894	0.1899	4
5136-00-821-1031	NO. 12-28UNF	TAPER	H2*	0.1933	0.1938	4
5136-00-825-0332	NO. 8-32UNC	TAPER	H2	0.1442	0.1447	4
5136-00-889-6610	NO. 2-56UNC	PLUG	H1	0.0744	0.0749	3
5136-00-889-6617	NO. 2-56UNC	BOTTOMING	H1	0.0744	0.0749	3
5136-00-889-6619	NO. 2-56UNC	TAPER	H1	0.0744	0.0749	3
5136-00-889-6629	NO. 2-64UNF	PLUG	H1*	0.0759	0.0764	3
5136-00-889-6987	NO. 2-64UNF	BOTTOMING	H1*	0.0759	0.0764	3
5136-00-889-6994	NO. 12-28UNF	TAPER	H3	0.1938	0.1943	4
5136-00-889-7011	0.094-48UNS*	PLUG	--	0.0808	0.0813	3
5136-00-966-5977	NO. 12-28UNF	PLUG	H3	0.1938	0.1943	4

* Exception to ANSI B94.9

Workmanship. All taps shall be free from imperfections which may affect their appearance, serviceability or durability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:09Z
Check the source to verify that this is the current version before use.

Quality assurance provisions. Sampling for inspection shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes, except where otherwise indicated herein. The lot size shall consist of all taps with the same NSN offered for acceptance at one time. A sample unit shall be one tap.

Visual and dimensional requirements. Each sample unit shall be examined and measured for compliance with applicable requirements of ANSI B94.9, and for workmanship as specified above. The inspection level shall be S-3 with an AQL of 1.5 percent defective.

Hardness test. Hardness testing shall be in accordance with ASTM E 18. The inspection level shall be S-2 with an AQL of 1.5 percent defective.

Performance test. When specified in the contract or purchase order, a sample consisting of three taps of the same manufacturing design shall be tested to demonstrate the ability of the tap to perform in various metals.

Each tap shall be tested by tapping ten through holes into one of three metal bars or plates, each composed of a different composition (one copper alloy, one carbon steel, and one corrosion resisting steel) of the suppliers option. The bars or plates shall have a minimum thickness equal to the major diameter of the tap, and the hole diameters shall be such as to produce a theoretical 75% thread height. Other test factors not specified shall be at the manufacturer's option. Failure of any of the three taps to produce smooth, well-defined Unified National Form threads of a size which is consistent from the first to the last tapped hole shall be considered cause for rejection of the entire preproduction sample.

Packaging inspection. An examination shall be made to determine compliance with all preservation, packaging, packing, labeling, and marking requirements as specified in the contract or purchase order. The inspection level shall be S-2 with an AQL of 2.5 defects per hundred units.

Regulatory requirements. In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes:
Copies of American National Standard ANSI B94.9, Taps, Cut and Ground Threads may be obtained from the American National Standards Institute, 1430Broadway, New York, New York 10018.
Copies of ASTM Standard E 18, Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.
Copies of Military Standards are available from the Naval Publication and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Air Force - 99

Review activity

Air Force - 84

PREPARING ACTIVITY:

GSA-FSS

Project 5136-0042

Source: https://assist.dla.mil -- Downloaded: 202640-26T08:09Z
Check the source to verify that this is the current version before use.